

Power board Natural-frequency comparison

20 undamped modes above 1 Hz | Six fasteners at 750 N each

Analysis date: 4 October 2026. ANSYS Mechanical/MAPDL 2026 R1.02.

The six bolts were tightened to 750 N each and locked. Static equilibrium and all three modal calculations completed with zero solver errors. Each modal case contains 20 real, undamped frequencies extracted above 1 Hz. The lowest retained frequency is 62.805 Hz for the original default model, 66.611 Hz for the refined default reference, and 485.964 Hz with the preloaded bolts. The default network contains a disconnected nine-body battery subassembly; the 1 Hz cutoff excludes its rigid-body motions. The comparison changes the fastening constraints and preload together.

Exploratory results: washer seating is assumed, the thin nickel strips retain poor element shapes, and the inherited battery/PCB material models are unvalidated. These limits matter despite numerical convergence.

Natural frequencies

Rank	Original default (Hz)	Refined default (Hz)	750 N bolts (Hz)
1	62.805	66.611	485.964
2	124.875	154.798	596.964
3	163.988	468.804	645.435
4	558.615	553.652	781.350
5	567.111	585.781	820.242
6	689.808	820.889	826.296
7	860.353	975.693	1122.018
8	945.371	1080.312	1139.112
9	1210.789	1128.582	1199.708
10	1281.230	1186.046	1366.076
11	1403.894	1346.580	1467.900
12	1522.247	1443.782	1714.134
13	1705.406	1701.164	1941.548
14	1744.806	1847.746	2003.039
15	1819.420	1873.051	2017.959
16	1894.293	1904.194	2135.663
17	2083.235	1969.429	2183.755
18	2116.840	2010.061	2212.560
19	2131.571	2222.743	2301.314
20	2205.547	2247.595	2364.179

Rows are frequencies in ascending order, not tracked equivalent mode shapes. Refinement also retains full integration and makes component-to-PCB contact direction explicit, so this is not an isolated mesh-convergence test.

Default-contact deformation

H: Modal

Mode 1 - normalized shape
 Type: Total Deformation
 Frequency: 66,611 Hz
 Unit: m
 2026. 10. 05. 15:06:05

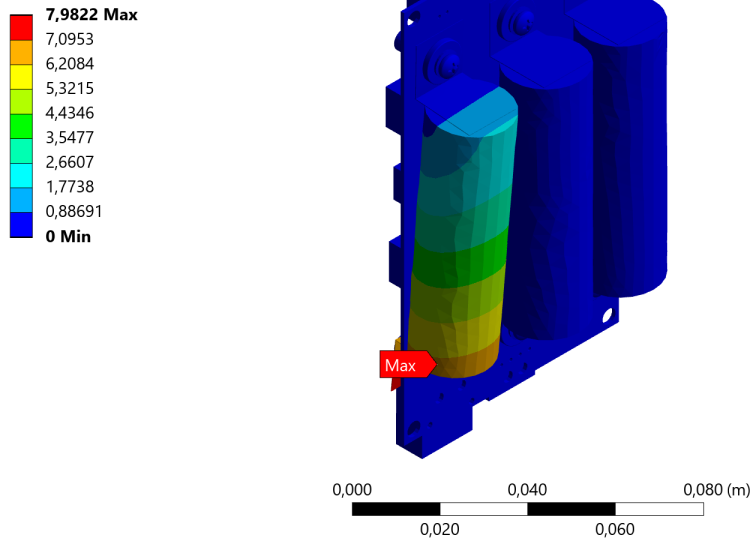


Figure 1: Refined default, rank 1: 66.611 Hz. The visible shape is dominated by battery and nickel-support motion; the PCB moves comparatively little.

H: Modal

Mode 20 - normalized shape
 Type: Total Deformation
 Frequency: 2247,6 Hz
 Unit: m
 2026. 10. 05. 15:06:07

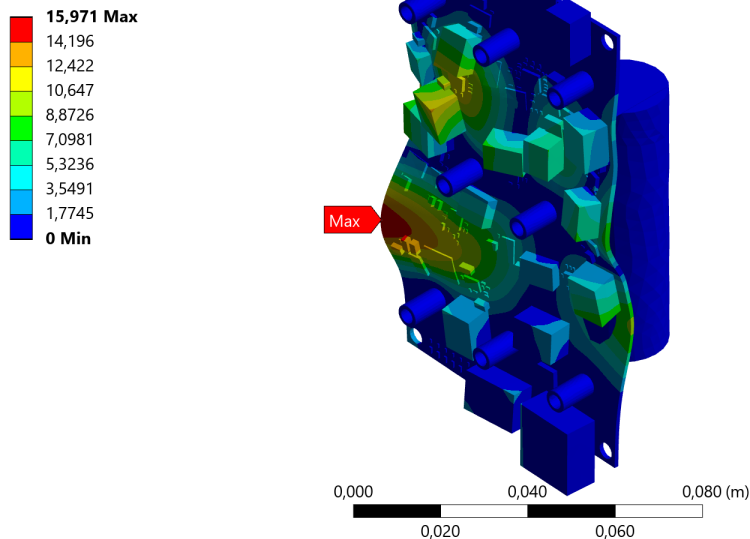


Figure 2: Refined default, rank 20: 2247.595 Hz. The PCB bends with pronounced motion along a long edge and its mounted components. Battery motion is comparatively small.

Mode-shape amplitudes are mass-normalized eigenvectors. Their displayed metre values and automatic scale factors are not physical displacement predictions. No excitation level was specified.

Six bolts at 750 N

The nonlinear static analysis converged at 750 N per bolt, then locked the six achieved tightening displacements. Final reactions were 749.997-749.999 N; tightening adjustments were 18.597-30.457 micrometres. Maximum static displacement was 0.04746 mm.

The modal solve uses the last locked static state (load step 2, substep 4). The results remain provisional because washer seating and the inherited material idealizations have not been confirmed against the physical assembly.

The first bolted mode (485.964 Hz) is dominated by motion at the lower PCB edge and its components. Modes 3 and 4 (645.435 and 781.350 Hz) include substantial battery motion. Mode 10 (1366.076 Hz) bends the upper edge; mode 20 shows broader PCB warping and coupled battery motion.

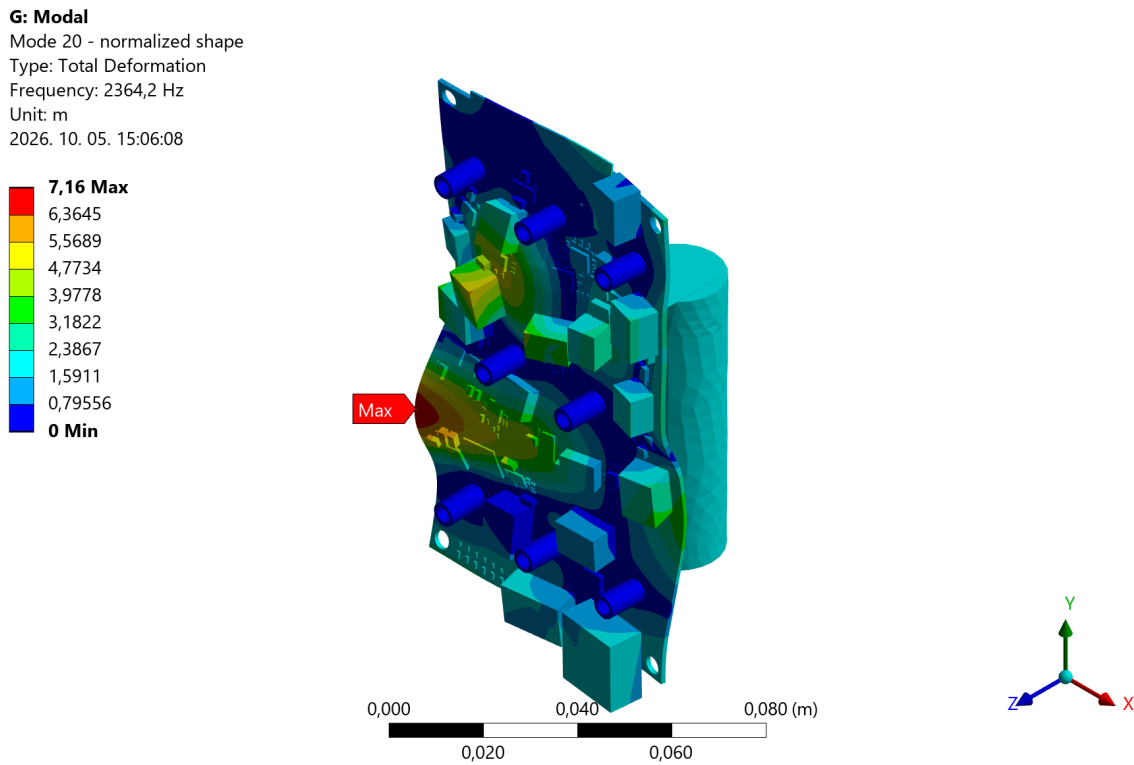


Figure 3: Six bolts at 750 N, rank 20: 2364.179 Hz. The PCB bends between the fixed stand-off locations, with coupled battery motion. This is an automatically scaled, mass-normalized mode shape; the legend is not a physical travel prediction.

Bolt location	Applied preload (N)	Locked reaction (N)
Left / upper	750.000	749.999
Centre / upper	750.000	749.999
Right / upper	750.000	749.999
Left / lower	750.000	749.997
Centre / lower	750.000	749.999
Right / lower	750.000	749.999

Model and numerical method

Common model and modal settings

ANSYS Mechanical/MAPDL 2026 R1.02, build 26.1, UP20260202, Windows 11 in Parallels. Solves use 12 distributed processes and consistent metre-kilogram-second-newton units. The assembled model has 208 bodies and approximately 0.25182 kg total mass.

Twenty real, undamped modes are requested with Block Lanczos over 1-10000 Hz. Eight existing stand-off end faces are fixed. The bolted modal analysis inherits supports, contact status and stress from the final locked static state.

Mesh and contact numerics

Original mesh: 95,215 nodes and 32,843 solid elements. Refined common mesh: 368,754 nodes and 112,902 solid elements; quadratic interpolation, 0.7 mm PCB sizing and 0.4 mm bolt sizing. No formal mesh convergence or experimental correlation has been established.

The solver originally reset full-integration SOLID186 elements to reduced integration. Full integration is retained in the refined analyses to avoid coarse-component hourglass modes. Existing component-to-PCB interfaces use explicit asymmetric MPC, with component faces as the constrained side. This numerical change also affects the spectrum.

Mesh-quality warnings remain in 1,498 tetrahedral elements, mostly in the thin nickel strips (maximum aspect ratio 216.3). These critical connections need improved meshes and a convergence study before the frequencies can be treated as validated predictions.

Bolt and battery idealizations

Six existing solid screws are cut with PRETS179 pretension sections. Each is loaded to 750 N in static step 1, then its achieved shortening is locked in step 2. Shaft-to-hole MPC ties bridge the CAD clearance; they idealize rigid fastening, not detailed threads or friction.

Provisional seating assumption: six metal washers bear against the PCB underside. Frictionless bearing contacts close the approximately 0.746 mm initial CAD gap through contact offsets. Four incidental washer-to-stand-off bonds are replaced in the bolted case. This seating arrangement remains to be confirmed against the real hardware. Batteries have no direct PCB contact or bond; their only connections are to nickel strips.

Interpretation of prestressed modes

The locked static equilibrium supplies the tangent stiffness for the modal calculation:

$$[\mathbf{K}_{\text{tan}}(\mathbf{u}_0, \boldsymbol{\sigma}_0) - \omega_i^2 \mathbf{M}] \boldsymbol{\phi}_i = \mathbf{0}, \quad f_i = \frac{\omega_i}{2\pi}.$$

Here $\boldsymbol{\phi}_i$ is a normalized mode shape, not the displacement produced by a specified excitation. Damping is disabled. A 1 Hz lower extraction bound filters lower-frequency modes; it does not physically restrain a disconnected assembly.

Qualification and traceability

Material and interpretation limits

Stored PCB properties are isotropic: $E = 23.330$ GPa, Poisson ratio = 0.15, density = 1969.78 kg/m³; thickness is 1.10236 mm. Stored battery properties are homogeneous: $E = 200$ GPa, Poisson ratio = 0.30, density = 2200 kg/m³. These inherited battery and PCB idealizations were not independently validated.

The requested comparison changes fastening constraints and prestress together. It does not isolate the effect of preload magnitude alone. Modal contour amplitudes are normalized, and a frequency cutoff does not establish overall constraint adequacy or structural stability.

Peak elastic static von Mises stress is 1.142 GPa; no local stress convergence or strength validation has been performed. Hardware MPC-overlap warnings also remain.

Accepted result evidence

Run evidence: `outputs/bolt_manager_work/resume_20261005_004228`. Accepted cases retain native inputs, solver logs, result files, frequency CSVs and Mechanical images. Input/result hashes and solver-log cross-checks are recorded in `accepted_results_audit.json`. Failed attempts are preserved separately and excluded.

Final static time: 2.0, load step 2, substep 4. Locked pilot reactions: 749.997-749.999 N. Independent washer bearing forces: 749.877-750.120 N; no contact damping force or excessive-penetration flag was recorded at the final state. Battery contact and shared-node audits passed.

The live project is left in the bolted configuration. Default results in this report come from preserved reference solves; rerunning a default analysis requires restoring its contact and bolt suppression settings.

Required validation before design use

Confirm washer seating and the actual load path against the physical assembly; improve the thin nickel-strip meshes and demonstrate frequency convergence; validate the PCB and battery material idealizations. Correlate important mode shapes with test data where available. Until these checks are complete, use the reported values as an exploratory comparison rather than qualified structural predictions.

References

1. [ANSYS: modal analysis and prestress workflow](#). ANSYS 2026 R1 documentation.
2. [ANSYS: ETCONTROL and automatic element-technology changes](#). ANSYS 2026 R1 documentation.
3. [ANSYS: singular matrices and hourglass modes](#). ANSYS 2026 R1 documentation.
4. [ANSYS: PRETS179 pretension element](#). ANSYS 2026 R1 documentation.

Reproduction procedure

This appendix records the accepted workflow and the files needed to repeat it. It was added on 5 October 2026 by checking the saved Mechanical commands, native input decks, solver logs and result audits. The existing numerical results have not been changed or re-solved for this documentation revision.

Starting point. Use a copy of the complete saved Workbench project and its companion project directory. The ECAD archive alone does not contain this assembled mechanical model, its added battery hardware, its inherited materials, or the established contact scopes. Exact numerical replay uses the supplied native decks; rebuilding from imported geometry can change entity IDs, mesh and results.

How to follow the appendix. Follow the numbered sections in order. Code is explicitly labelled as Mechanical IronPython, Windows PowerShell, MAPDL, or report-build commands. Replace only the stated paths and fresh output names. Retain the accepted evidence folders. The code excerpts are instructions reconstructed from executed commands, not a newly tested one-click installer. Full original helper scripts are supplied as references; some contain historical paths, global variables, or one-run guards and must not be launched blindly.

The source bundle contains a **reproduction/** directory with native inputs, contact/body inventories, custom bolt-library files, selected original scripts, acceptance evidence and a SHA-256 manifest. Large binary solution/restart files and the complete CAD/Workbench project are retained separately in the project tree, not duplicated in the report ZIP. Their dependencies are identified below.

Procedure index

1. Preserve the project and establish the environment

1. Close or save Mechanical, then copy the entire project folder to a new study folder. Keep `powerboardsim_v1.wbpj`, `powerboardsim_v1_files/`, and `outputs/` together. Open the copied project in Workbench and open its Model cell in Mechanical. Do not substitute a lone old `SYS.mechdb` into the current project: saved model files are snapshots, not complete portable projects.
2. Use ANSYS Mechanical/MAPDL 2026 R1.02, build 26.1 UP20260202. The accepted solver used 12 distributed processes. The original VM was Windows 11 in Parallels with 14 virtual CPUs and 112 GB RAM. Keep enough local disk space for per-rank matrices and results: one random-case staging copy was approximately 25 GB before new results.
3. Load the existing APDL Bolt Manager extension (internal extension name `BoltTools`, version 2024.1). Sherlock's imported geometry and material assignments must already be present. The accepted export also identifies Sherlock_App 2026.1. Use solver units m, kg, s, N; the geometry and mesh sizing tables below are in mm.
4. Establish a new output folder. In the original project, call the project root P, the accepted modal run B, and the accepted random run R, using the paths below. A fresh rerun must use different output directories. Do not reuse completion markers from an old run as evidence that a new run passed.

```
# Original Windows project root (Z: maps the macOS user home):
P = r'Z:\Developer\MIT_Rkt_Team\2026-7\MIT-RT-26_27-Work\simulation_work\avionics\powerboardsim\1'
B = P + r'\outputs\bolt_manager_work\resume_20261005_004228'
R = P + r'\outputs\random_vibration_20261005_0950'
# Native solver executable:
# C:\Program Files\ANSYS Inc\v261\ansys\bin\winx64\ANSYS261.exe
```

Identify analyses by their role, then verify their IDs

Role in the saved model	Object ID	Native directory under dp0
Static, 750 N load then lock	4123	SYS-2/MECH
20-mode prestressed comparison	1656	SYS-1/MECH
Default comparison, later refined	4182	SYS-3/MECH
Expanded modal basis for random	4254	SYS-4/MECH
Random X / Workbench block F	4259	SYS-5/MECH
Random Y / Workbench block E	4264	SYS-6/MECH
Random Z / Workbench block D	4269	SYS-7/MECH

Object IDs, Workbench letters and folder names can change when objects are recreated. A new study should resolve the corresponding objects and confirm their names, types and upstream links before using any ID from this report. A geometry ID is not a Mechanical object ID.

Open Mechanical's Scripting pane and select its IronPython environment for the snippets below. Use `System.IO` for files there. The previous `ImportError: No module named _json` is avoided by keeping JSON parsing in host Python 3; it is not repaired by changing the material model.

2. Verify the complete assembly, supports and materials

1. Expand Geometry. Verify 208 unsuppressed, meshed bodies, including 172 imported `COMP_` electronic-component bodies, three cell bodies, nickel strips, six battery screws, washers and stand-offs. Keep component solids in the solve even when making a PCB-only contour. The assembled mass is approximately 0.25182 kg.
2. Check `reproduction/common/assembly_body_inventory.tsv`: it lists each body's geometry ID, Mechanical object ID, material and meshed node/element counts. Match the inventory before altering contacts. PCB geometry ID is 42342 (Mechanical body object 3558); the PCB measures approximately $69.215 \times 120.777 \times 1.10236$ mm.
3. Scope one Fixed Support to the eight existing mounting end faces below. In the accepted static analysis its object ID is 4127; the default analysis uses a separate support on the same faces. Do not fix the six battery screws directly, and do not add weak springs to hide a disconnected assembly.
4. Retain the inherited material assignments. Verify elastic stiffness and density in Engineering Data and in the exported native material records. Assigning one generic material to all components would create a different study. The full native decks contain the actual material definitions and their element assignments; the table below only highlights the large structural bodies.

Scope	Geometry entity IDs
Fixed mounting faces	23075, 23135, 23195, 23255, 23315, 23375, 23435, 23495
PCB / underside seat target	Body 42342 / face 41909
Battery cells	20157, 20675, 20822
Nickel strips	20091, 20297, 20348, 20642, 20756, 20929

Stored material	E (Pa)	ν	ρ (kg/m ³)
PCB	23330241935.4839	0.15	1969.77741935484
PilesLiPo	2.0×10^{11}	0.30	2200
NICKEL-200	2.2×10^{11}	0.305	8874
Acier standard	2.0×10^{11}	0.30	7850

Initialize the Mechanical API without relying on a previous console session

```
import System
from Ansys.Core.Units import Quantity
from Ansys.ACT.Interfaces.Common import SelectionTypeEnum
from Ansys.Mechanical.DataModel.Enums import *
m = ExtAPI.DataModel.Project.Model
static = ExtAPI.DataModel.GetObjectById(4123)
bolted = ExtAPI.DataModel.GetObjectById(1656)
ext = [x for x in ExtAPI.ExtensionManager.Extensions
       if x.Name == 'BoltTools'][0]
manager = [x for x in ExtAPI.DataModel.GetUserObjects(ext)
           if x.Name == 'ApdlBoltManager'][0]
bm = ext.GetModule().ApdlBoltModule
def sel(ids):
    s = ExtAPI.SelectionManager.CreateSelectionInfo(
        SelectionTypeEnum.GeometryEntities)
    s.Ids = list(ids)
    return s
def enumset(obj, name, value):
    p = obj.GetType().GetProperty(name)
    p.SetValue(obj, System.Enum.Parse(p.PropertyType, value), None)
```

3. Restore the intended contacts and refined mesh

1. Preserve the existing contact definitions and scopes. Do not run automatic contact generation as a re-production step: that was previously capable of changing pairings. The accepted bolted network has 210 active contact regions: 172 component-to-PCB bonds, 8 stand-off-to-PCB bonds, 24 hardware/nickel/battery bonds, and 6 frictionless bearing seats. Additional shaft-to-hole MPC elements are generated by the bolt extension.
2. Enforce the user's battery rule: only battery-to-nickel connections are active. The six allowed contact regions are listed below. Suppress any battery-to-PCB, battery-to-washer or other direct battery contact. Do not replace them with frictionless battery-to-PCB contact; even that would introduce an extra load path. Check for shared mesh nodes as well as contact objects.
3. For the refined models, set the 172 component-to-PCB bonds to Asymmetric, MPC, Nodal Normal to Target. Component faces are the contact/source side; PCB faces are the target side. Retain the detailed face scopes from the saved project. `reproduction/common/final_active_contacts.tsv` records body-level pairs; the native decks preserve the exact meshed contact surfaces.
4. In Mesh, retain existing controls and set Element Order to Quadratic. Add or edit body sizing: PCB body 42342 at 0.7 mm; screw bodies 21279, 21629, 21979, 22329, 22679, 23029 at 0.4 mm. Generate the mesh. Expected geometry mesh: 368754 nodes and 112902 solid elements; PCB alone: 217704 nodes and 39014 elements. Remeshing in a different release can change this mesh. Use the archived decks for identical node/element replay.
5. Insert the following Commands object into the static analysis and the refined-default modal analysis. Keep legacy Commands object 4113 suppressed. Preserve the export order so that the command executes before SOLVE. The original unrefined default did not receive this numerical correction.

```
FINISH
/PREP7
ETCONTROL,OFF
SHPP,WARN
FINISH
/SOLU
```

Contact object	Source body	Target body	Accepted type
1699	20091	20157	Bonded
1711	20157	20348	Bonded
1726	20297	20675	Bonded
1753	20642	20675	Bonded
1759	20756	20822	Bonded
1768	20822	20929	Bonded

In Mechanical, set PCB670_RUN to a fresh audit directory and run the supplied script `reproduction/common/audit_battery_constraints.py`. It checks the six allowed contact pairs and rejects shared nodes between batteries and any non-nickel body. It writes a contact inventory; it does not solve.

The accepted mesh still has 1498 tetrahedra with shape warnings, mostly in nickel, with maximum reported aspect ratio 216.3. Reproducing those warnings is traceability, not a validation target. Mesh improvement would be a new study. The refined comparison also changes integration/contact numerics and therefore is not a pure mesh-convergence experiment.

4. Configure the six APDL Bolt Manager definitions

1. Use the six existing flexible solid screws. Keep the saved APDL Bolt Manager and its six children; do not add six additional geometric screws or duplicate pretension objects. Child IDs in the accepted project are 4139, 4147, 4155, 4163, 4171 and 4179. The manager ID is 4122.
2. Make the three files in `reproduction/custom_library/` available in BoltTools' configured custom-part library and reload it with `bm.LoadXmlApdlBolts()`. Check that `bm.ApdlBoltDict['PCB_ExistingBolt_670N.xml']`. `LoadErrors` is empty. The old 670N filename is retained for compatibility; its **actual stored load is 750 N**. Do not change it back to 670 N.
3. For each row below, create or verify three geometry Named Selections: `PCB670_nn_BODY`, `PCB670_nn_SHAFT`, `PCB670_nn_HOLE`. Send them to the solver as node components. Use the reported body and faces, not all shaft-adjacent parts. The exported body component is selected as elements in the custom preprocessor commands.
4. Verify a global-aligned coordinate system at each listed x, y and $z = -3.772678375$ mm. On each Bolt Manager child, select that one coordinate system; set the bolt part to the custom XML; set Bolt Washer, Nut and Nut Washer to None; set Display/Visible to No. The custom definition adds pretension/contact elements to existing solids rather than extra bolt mass.

No.	Position	Body	Shaft faces	PCB hole faces (inclusive)
1	Left, high Y	21279	21214, 21273	41970–41977
2	Centre, high Y	21629	21564, 21623	41962–41969
3	Right, high Y	21979	21914, 21973	41954–41961
4	Left, low Y	22329	22264, 22323	41914–41921
5	Centre, low Y	22679	22614, 22673	41922–41929
6	Right, low Y	23029	22964, 23023	41930–41937

No.	x (mm)	y (mm)
1	10.81234895	102.90699896
2	34.81234887	102.90699896
3	58.81234883	102.90699896
4	10.81234894	17.90699886
5	34.81234886	17.90699886
6	58.81234882	17.90699886

Each child's User Commands entry is sent at Assembly level for the Bolt part, in both Pre and Solve phases. For bolt 1, use the following text; change the index and the three solver names for each successive bolt.

```
PBIndex=1
PBBody='PCB670_01_BODY'
PBShaft='PCB670_01_SHAFT'
PBHole='PCB670_01_HOLE'
```

The API representation is `obj.Attributes['UserCmds']` containing `[['Existing solid bolt scopes', text, ['Pre', 'Solve'], ['Bolt'], ['Assembly']]]`. After editing, call `obj.Controller.GetData(obj, False)` and `obj.NotifyChange()`. The bundled historical setup script shows the full API construction, but its old 670 N assumptions and empty-manager checks must not be replayed on the final project.

5. Define the seats and the load–lock sequence

1. For the bolted configuration, unsuppress all six Bolt Manager children. Suppress four incidental washer/stand-off bonds: 1696, 1732, 1744, 1771. Create or unsuppress the six bearing contacts below, with washer as contact/source and PCB underside 41909 as target.
2. Set each bearing contact to Frictionless, Asymmetric, Augmented Lagrange, Pinball Region = Radius, Pinball Radius = 1 mm, Interface Treatment = Adjust to Touch. This deliberately closes the approximately 0.7459997 mm CAD gap. It is the accepted seating assumption; confirm its physical validity separately.
3. In Static Structural, use two steps ending at 1 s and 2 s. Set Large Deflection = On, Weak Springs = Off, Automatic Time Stepping = On for both steps, Initial Substeps = 5, Minimum = 1, Maximum = 100. These times index a quasi-static loading sequence; they are not a vibration duration. Preserve restart data for the downstream modal calculation.
4. The custom XML uses global cut plane PBCutZ=-2 mm, upper shaft range PBTopZ=-1 mm to PBMaxZ=2 mm, and MPC pinball PBPInb=1.5 mm. PCB_ExistingBolt_Pre.inp uses PSMESH/PRETS179 and TARGE170/CONTA174. It reserves 2000 node and element IDs per bolt and tests that the cut, target and contact selections are nonempty. Keep the entire supplied file unchanged for exact replay.

Seat	Contact object	Washer body	Washer face	PCB face
1	4224	20246	20234	41909
2	4228	20040	20028	41909
3	4232	20525	20513	41909
4	4236	20466	20454	41909
5	4240	20878	20866	41909
6	4244	20407	20395	41909

Exact load-phase commands in the custom library

The following is the operative content of PCB_ExistingBolt_Load.inp. PBLoad is supplied as 750 N by the XML. The analysis-type guard prevents applying static tightening again during modal or PSD solves.

```
! Static step 1: tensile preload. Step 2: lock achieved adjustment.
! Prestressed modal obtains the locked state from its linked static solution.
*GET,PBAnalysis,ACTIVE,0,ANTY
*IF,PBAnalysis,EQ,0,THEN
PBPN=PBPIlot%PBIndex%
*IF,_MechAnalysisStep,EQ,1,THEN
F,PBPN,FX,PBLoad
/COM,PCB_BOLT_%PBIndex% PRELOAD %PBLoad% N
*ELSEIF,_MechAnalysisStep,GT,1,THEN
FDELE,PBPN,FX
D,PBPN,UX,%_FIX%
/COM,PCB_BOLT_%PBIndex% LOCK
*ENDIF
*ENDIF
ALLSEL,ALL
```

The UX degree of freedom here belongs to a PRETS179 pretension pilot and represents relative tightening, not an instruction to move the physical screw in global X. Step 1 applies force; step 2 removes that force and locks the achieved adjustment with %_FIX%. Keep this sequence; leaving 750 N force-controlled in step 2 would be a different tangent-state model.

6. Preflight, solve and verify the static parent

1. In Mechanical, export a fresh static native deck after refreshing the ACT writer. Use a new folder. The commands below assume `newout` is an existing Windows directory and that section 2 initialized the API objects.

```
bm.StartApdlInputFileWrite(static)
static.WriteInputFile(newout + r'\static_input.dat')
s = System.IO.File.ReadAllText(newout + r'\static_input.dat')
assert s.count('PBLoad = 750.0') == 12 # six bolts, two steps
assert s.count('CMSEL,S,%PBBody%,ELEM') == 6
assert 'ETCONTROL,OFF' in s
assert '*GET,PBAnalysis,ACTIVE,0,SOLU,ANTYPE' not in s
```

2. Run a separate no-SOLVE preflight on a copy of the exported input, truncated immediately before its first standalone SOLVE command. The supplied `reproduction/native/static750/preflight.dat` is the exact accepted preflight deck and ends in inventories, `CNCHECK,DETAIL` and `/EXIT,NOSAVE`. Use it unchanged only with its corresponding mesh/deck. For changed geometry, regenerate the preflight prefix rather than transplanting old mesh records.
3. Launch MAPDL in a fresh directory with `-b nolist -smp -np 1 -p preppost -j check -i preflight.dat -o preflight.out`. Require zero native errors, completion of the preflight, six nonempty pretension cuts, and 750 N on each pilot. The accepted bolt inventories are below. Also pass the nickel-only battery audit.
4. Re-export `launch_static_input.dat` immediately before solving. Compare all non-comment records with the checked export; tolerate export timestamp comments only. If anything substantive differs, repeat the preflight. Then solve Static Structural using 12 distributed processes (Mechanical `static.Solve(True)` was used).
5. Wait for the solve to finish before any next command. Check Mechanical Solution = Solved and the native `solve.out`: zero errors and final convergence at time 2. The accepted final state was load step 2, substep 4. Read the pretension audit at both steps and verify all six final locked reactions within 1 N of 750 N.

Bolt	PRETS179	TARGE170	CONTA174	Locked force (N)
1	142	187	32	749.999347595
2	141	200	32	749.999448517
3	149	200	32	749.999081634
4	134	180	32	749.996796406
5	138	185	32	749.999422828
6	126	185	32	749.998678460

Preserve the restart and equilibrium evidence

Archive the full static solver directory, including per-rank restart data, `file.rdb`, `file.ldhi`, restart substep files, `file.rst`, `ds.dat`, `solve.out`, the error file and bolt audit. A result file alone is insufficient for the perturbation modal restart. Do not clear generated data between static and modal.

Accepted static evidence is `B/bolted750_refined/`; `B` denotes the root defined in section 1. The similarly named `bolted750`, `bolted750_asymmetric` and `bolted750_fullintegration` directories contain earlier attempts and are not the final parent. Peak static displacement was 0.0474595 mm; peak elastic equivalent stress was approximately 1.142 GPa. Final washer bearing forces were 749.877–750.120 N, with zero reported contact damping force. Those stresses remain unqualified.

7. Extract and check the tightening state

Insert this Commands object under the static Solution before solving, or run equivalent MAPDL postprocessing on the preserved static result with its parent database loaded. It is the actual force-check procedure. It writes the relative adjustment and reaction at all six pretension pilots at the end of each step.

```
/POST1
*CFOPEN,bolt_preload_audit,csv
*VWRITE
('step,bolt,pilot,adjustment_m,reaction_N')
*DO,PBSTEP,1,2
SET,PBSTEP,LAST
*DO,PBII,1,6
PBPB=PBPilot%PBII%
*GET,PBUX,NODE,PBPN,U,X
*GET,PBRF,NODE,PBPN,RF,FX
*VWRITE,PBSTEP,PBII,PBPB,PBUX,PBRF
(F3.0,' ',F3.0,' ',F10.0,' ',E20.12,' ',E20.12)
*ENDDO
*ENDDO
*CFCLOSE
ALLSEL,ALL
```

Bolt	Accepted pilot node	Locked adjustment (m)
1	369771	2.82970332153E-5
2	371772	2.22045874625E-5
3	373773	1.85968938693E-5
4	375774	3.04568660374E-5
5	377775	2.06860396937E-5
6	379776	1.87719835696E-5

Pilot IDs can change if the mesh or object ordering changes. The APDL parameters `PBPilot1` through `PBPilot6`, created by the bolt library, locate the current pilots. Do not hard-code the table's pilot numbers into a remeshed model.

Step-1 pilot reaction can be nearly zero because its pretension degree of freedom is force-loaded rather than displacement-constrained. That value is not evidence of missing preload. Check the applied 750 N in the input/preflight and the final step-2 locked reaction. The load is 750 N *per bolt*, not 750 N divided among six.

For an independent check, inspect Contact Tool/bearing output at the final locked substep and compare the six compressive seat forces with the archived `reproduction/common/bearing_contact_audit.json`. A plausible pilot force alone does not prove that the washer-to-PCB load path is correct.

Before modal extraction, confirm that all six bolt definitions remain active, the six seats are active, the four incidental bonds remain suppressed, the eight supports are unchanged, and the battery audit still passes. Save the project. Do not accidentally restore default contacts while calculating a bolted modal or random case.

8. Reproduce both default-contact references

The original default and refined default are distinct references. They were solved *before* activating the final bolt/washer model and were archived before the shared Mechanical mesh or contacts changed. The live analysis 4182 now corresponds to the refined reference; its current tree is not a way to regenerate the original coarse mesh by pressing Solve.

1. For exact original-default numerical replay, use `reproduction/native/default_original/ds.dat`. It contains the original mesh, materials, contacts, support-node sets and 1–10000 Hz modal request. The accepted original mesh has 95215 geometry nodes and 32843 solid elements. Retain its archived preflight and solve log for comparison.
2. To reproduce the refined-default model in a copied project, first apply the common refined mesh and component-contact numerics in sections 2–3. Suppress all six APDL Bolt Manager children. Suppress bearing contacts 4224, 4228, 4232, 4236, 4240 and 4244 if they already exist. Restore the original incidental bonds 1696, 1732, 1744 and 1771. Keep the six battery-to-nickel bonds and existing component/stand-off connections. Keep Commands object 4113 suppressed.
3. Use a separate, unprestressed Modal analysis with a Fixed Support on the same eight mounting faces. Do not link it to static 4123. Set Max Modes to Find = 20, Limit Search to Range = Yes, minimum = 1 Hz, maximum = 10000 Hz, Damped = No, solver = Program Controlled. Verify native `MODOPT,LANB,20,1.,10000..` The accepted final solver was Block Lanczos; an earlier 100000000 Hz trial is not the reported case.
4. Preserve full integration for the refined default with the section-3 Commands object. Export, run the no-SOLVE contact/mesh preflight, compare the launch export with the checked deck, and solve with 12 distributed processes. Require zero solver errors and 20 real frequencies, all at least 1 Hz.
5. Archive `ds.dat`, `solve.out`, `file.rst`, the error file, frequency CSV and images immediately, before restoring the bolted contact state. Use separate directories for coarse and refined results.

Mechanical settings excerpt for the refined default

```
baseline = ExtAPI.DataModel.GetObjectById(4182)
for obj in list(manager.Children): obj.Suppessed = True
for oid in [4224,4228,4232,4236,4240,4244]:
    ExtAPI.DataModel.GetObjectById(oid).Suppressed = True
for oid in [1696,1732,1744,1771]:
    ExtAPI.DataModel.GetObjectById(oid).Suppressed = False
aset = baseline.AnalysisSettings
aset.MaximumModesToFind = 20
aset.Damped = False
aset.LimitSearchToRange = True
aset.SearchRangeMinimum = Quantity('1 [Hz]')
aset.SearchRangeMaximum = Quantity('10000 [Hz]')
aset.SolverType = SolverType.ProgramControlled
```

The original default has a disconnected nine-body left-battery subassembly: 20091, 20157, 20246, 20348, 20466, 20561, 20708, 21279, 22329. The 1 Hz lower bound omits its rigid-body modes; it does not connect those bodies. Adding a constraint to eliminate that mechanism would change the baseline.

9. Solve the 20-mode bolted comparison

1. Restore the bolted contact state from section 5: bolts and six seats active, four incidental bonds suppressed. Re-run the static solve if its state was invalidated by changing the shared mesh or contact definitions. Require the checks in sections 6–7 to pass for the state being used now.
2. Set the bolted Modal analysis' Pre-Stress source to static 4123 and use its final locked state. In the preserved tree this initial-condition object is 1659. Verify the source explicitly rather than relying on the system's displayed letter. The accepted modal starts from static time 2, load step 2, substep 4.
3. Set 20 modes, 1–10000 Hz, Damped = No. Preserve the existing contact-splitting and retained-file settings from the saved model. Export after `bm.StartApdlInputFileWrite(bolted)`. The native deck must be a perturbation restart, without another mesh, another PSMESSH operation, or a repeated static tightening load.
4. Preflight the exported modal guards against a copy of the converged parent `file.rdb`, without SOLVE. Check that all six pretension element sets exist and that the analysis type is modal (`ANTY = 2`). The accepted check is bundled as `reproduction/native/modal20/preflight.dat`; its absolute RESUME path must refer to the copied static parent. Verify the final static reactions separately, since a database-only check is not a result check.
5. Re-export and compare the non-comment input records. Solve the modal analysis with 12 distributed processes, allow Mechanical to transfer the complete static restart dependencies, and wait for completion. Require zero native errors and exactly 20 real modes. Cross-check the result-reader frequencies against the eigenvalue table in `solve.out`.

```
# Mechanical, on the preserved project IDs:
ic = ExtAPI.DataModel.GetObjectById(1659)
ic.PreStressICEnvironment = static
assert ic.PreStressICEnvironment.ObjectId == static.ObjectId
aset = bolted.AnalysisSettings
aset.MaximumModesToFind = 20
aset.Damped = False
aset.LimitSearchToRange = True
aset.SearchRangeMinimum = Quantity('1 [Hz]')
aset.SearchRangeMaximum = Quantity('10000 [Hz]')
bm.StartApdlInputFileWrite(bolted)
bolted.WriteInputFile(newout + r'\modal_input.dat')
# Only after the documented preflight and export comparison:
bolted.Solve(True)
```

Native commands that identify the accepted prestressed extraction

```
/ASSIGN,RSTP,file,rst
/SOLU
ANTYPE,,RESTART,,PERTURBATION
PERTURB,MODAL,,CURRENT,DZEROKEEP
SOLVE,ELFORM
MODOPT,LANB,20,1.,10000.
! Keep the generated output controls and guarded ACT load blocks.
SOLVE
```

This excerpt is an identification check, not a standalone replacement deck. The short modal input depends on the complete static restart data, including per-rank files. Loading only `file.rst`, only `file.rdb`, or a clean empty solver directory will not reproduce the perturbation restart. Use the complete archived parent or regenerate static first.

10. Export modal frequencies and deformation figures

1. Once an analysis is Solved, obtain its frequencies from `GetResultsData().ListTimeFreq`. Do not read the displayed zero-frequency result placeholder or infer eigenvalues from the modal deformation legend. The snippet below creates a CSV in a fresh export directory.
2. Under each modal Solution, create Total Deformation results with Mode = 1, 2, 3, 4, 10 and 20 (reuse existing results if present). Evaluate All Results. Use automatic deformation scaling, all bodies, no mesh lines, white background and an isometric view. The reported contour amplitudes remain normalized eigenvectors.
3. Use `reproduction/scripts/export_modal_results.py` for the original export procedure. Set `PCB670_RUN` to the new export root, `REPORT_CASE` to the case subdirectory and `REPORT_ANALYSIS_ID` to the current analysis ID before running it in Mechanical. Create that subdirectory first. The script also archives the native result and log; allow that copy to finish.

```
analysis = bolted # or the separately verified default analysis
assert unicode(analysis.Solution.ObjectState) == 'Solved'
reader = analysis.GetResultsData()
freqs = list(reader.ListTimeFreq)
reader.Dispose()
assert len(freqs) == 20 and all(f >= 1.0 for f in freqs)
System.IO.File.WriteAllText(newout + r'\frequencies.csv',
    'mode,frequency_hz\n' + '\n'.join(
        '%s,%.12g' % (i+1,f) for i,f in enumerate(freqs)))
```

Views used in this report

The refined-default first mode uses the battery-facing view: `ViewVector = (1, 1, -1)` and `UpVector = (-1, 2, 1)`, then Fit. The refined-default twentieth mode and bolted twentieth mode use the isometric view, Fit. Native PNG exports use 1800 by 1200 pixels, a white background, `CurrentGraphicsDisplay = False`, `ShowMesh = False` and `ExtraModelDisplay = NoWireframe`. Camera fitting may vary slightly with the viewport; the scalar result and mode index determine numerical reproducibility.

Acceptance checkpoints

Case	Modes	First (Hz)	Twentieth (Hz)
Original default	20	62.8049504974	2205.54668797
Refined default	20	66.6112548974	2247.59456156
Six 750 N bolts	20	485.964268212	2364.17911966

Compare all 20 rows with the supplied CSVs, not just these endpoints. For an unchanged native mesh/deck and solver, investigate any difference beyond export rounding. A regenerated mesh, a changed contact state, a changed integration option or a different material set can legitimately produce a different spectrum and should be reported as a new analysis. No engineering acceptance tolerance was established by this study.

Use `reproduction/common/accepted_results_audit.json` to check the original input, log, result-file and frequency hashes. Rerun binaries can differ in bookkeeping and floating-point details, so hashes identify the original evidence rather than guaranteeing identical newly generated binaries. Compare numerical values and solver state as well as provenance.

11. Native replay, source files and report rebuilding

For an exact archived default mesh, a full native deck can be solved directly without regenerating CAD or the mesh. Copy the selected `ds.dat` to a *new* Windows directory as `run.dat`. Review any absolute Workbench scratch/output paths and redirect those bookkeeping destinations into the new run; retain all physics and node/element records. Run its no-SOLVE preflight first. The static full deck is similarly self-contained for model construction, but its downstream modal deck is a restart job and requires the static dependencies described above.

Windows PowerShell, fresh folder, after preflight:

```
$solver = 'C:\Program Files\ANSYS Inc\v261\ansys\bin\winx64\ANSYS261.exe'
$runDir = 'C:\Temp\PCB_REPLAY_DEFAULT_20261005_01'
# Populate this new directory with the reviewed run.dat first.
Get-FileHash (Join-Path $runDir 'run.dat') -Algorithm SHA256
$job = Start-Process $solver -WorkingDirectory $runDir -PassThru `
  -ArgumentList '-b nolist -s noread -dis -np 12 -p ansys -j file -i run.dat -o solve.out'
# Wait for this job and its solver ranks to finish, then inspect solve.out.
```

The original VM had a single `ansys` solver seat. If an external solve cannot obtain it, save the project and release Mechanical's held license with `ExtAPI.Application.LicensePreference.DeActivateLicense()`. Run one solve at a time. Reacquire `Ansys Mechanical Enterprise` before Mechanical result evaluation. Do not start duplicate jobs to work around a license wait.

Files shipped in the report source bundle

Under reproduction/	Purpose
<code>native/default_original/</code>	Original default deck, no-SOLVE preflight, log, frequency CSV.
<code>native/default_refined/</code>	Refined default deck, preflight, log, frequency CSV.
<code>native/static750/</code>	Accepted static deck, preflight, force inventory and log.
<code>native/modal20/</code>	20-mode restart deck, parent-check preflight, log and frequencies.
<code>custom_library/</code>	Actual 750 N XML and both APDL source files.
<code>common/, scripts/</code>	Body/contact inventories, audits, original implementation/export references.
<code>MANIFEST.json</code>	Source paths, byte counts and SHA-256 for every bundled evidence file.

Original project backups are under `B/default/`, `B/before_mesh_refinement/` and `R/project_before_random/`. These individual project/model snapshots are useful evidence but do not replace a complete project directory or solver restart archive.

To rebuild this report from its source ZIP, extract the whole ZIP, change into the extracted report directory, and run the two commands below using a TeX distribution with pdfLaTeX and the packages in the preamble. The reproduction appendix is directly in the source. The existing three modal image payloads remain embedded in this original report format.

```
pdflatex -interaction=nonstopmode -halt-on-error power_board_modal_comparison.tex
pdflatex -interaction=nonstopmode -halt-on-error power_board_modal_comparison.tex
```

The first pass builds cross-references and the appendix contents; the second resolves them. Check the log for errors and clipped boxes, then inspect the PDF pages. Do not run an old report-generation script over the revised source: those historical builders predate this appendix.

English figure refresh: 5 October 2026

All report contours were re-exported from Mechanical after changing its interface language to English. To reproduce the heading translation, rename the relevant Workbench system titles to **Modal** or **Random Vibration**; changing the interface language alone leaves saved system titles unchanged. The supplied `reproduction/scripts/translate_workbench_titles.py` performs these display-name changes through the Workbench journal API. Adapt its project-specific system names and audit path before reuse.

Then run `reproduction/scripts/export_english_figures.py` in Mechanical, with `OUT` changed to a fresh existing directory and all result IDs checked against the saved project. The script validates the captions, result maxima and the retained modal frequencies, then exports PNGs without solving. The refined-default modal branch retains the accepted baseline results but is marked obsolete because the shared model now uses the bolt configuration; it was not recalculated against those changed contacts. The export audit and heading-translation audit are in `reproduction/common/english_figures/`. Numerical results, result scopes and documented camera directions are unchanged. The straight Back view uses a scene-height multiplier of 1.25 after Fit to keep the complete deformed outline clear of the image edges.

12. Reproduce the companion 4K modal videos

The three companion MP4s correspond to the report’s refined-default modes 1 and 20, and the six-bolt mode 20. All use native Mechanical geometry and contours at 3840 by 2160 pixels, encoded at 60 frames per second. They show normalized eigenvectors with deliberately enlarged, slowed motion. Their amplitudes and playback periods are not physical predictions.

File stem	Result	Frequency (Hz)
<code>modal_fig01_default_mode1</code>	Refined default, mode 1	66.6112548974
<code>modal_fig02_default_mode20</code>	Refined default, mode 20	2247.59456156
<code>modal_fig03_bolted_mode20</code>	Six 750 N bolts, mode 20	2364.17911966

A. Preserved default-mode animations

1. Open the saved project and select the retained refined-default result. Check its mode number and frequency against the table. The shared model now contains bolt contacts; do not solve or regenerate the obsolete default branch merely to export an animation. Re-solving would create a different model, rather than reproduce this comparison.
2. Set an English result/system title, show the complete assembly, hide mesh lines, and use the documented battery-facing or isometric camera. Fix the contour range throughout the animation.
3. The original export source in `reproduction/videos/` sets the animation options to `NumberOfFrames = 91`, `Duration = 1.5 s`, `UpdateContourRangeAtEachFrame = False`, and `FitDeformationScalingToAnimation = True`. It calls `ExportAnimation` with `AnimationExportSettings(3840, 2160)` and a separate `TemporaryFramesPath`. Retain the actual exported image sequence: the accepted full cycle contains 180 native 4K PNG frames. Validate their count and dimensions instead of assuming that the option value equals the final full-cycle frame count.
4. Preserve the native legend separately. Encode the 180-frame cycle three times at 60 fps, producing a nine-second MP4 with three visually slowed cycles. The native graphics are not upscaled. The encoder overlays an English caption identifying normalized motion and three seconds per displayed cycle.

B. Bolted mode 20: explicit phase export

For this final clip, use explicit PNG exports to avoid the native video-encoding stage, which stalled during earlier export attempts. This changes only post-processing, not the accepted modal calculation.

1. Copy the accepted bolted mode-20 result into a fresh read-only post-processing directory. Its archived `file.rst` has 247332864 bytes and SHA-256 `812cacda84f169b54920ccb8b971210523df45b4aa0cad17feaac050b8a8ed9d`. Do not use a transient result, a default-contact result, or a failed trial.
2. Read result set 20 with ANSYS DPF displacement operator `U`. Confirm the frequency 2364.17911965844 Hz through Mechanical’s native result reader. The maximum normalized displacement magnitude is 7.160033972903191 in the retained eigenvector normalization. Seven checkpoint vectors are supplied with the export audit.
3. Compute a constant contour using `norm_fc`. For frame $i = 0, \dots, 179$, compute a warp field with `scale_fc`:

$$\mathbf{u}_{\text{warp},i} = \sin(2\pi i/180)\phi_{20}.$$

Set the Python Result workflow’s contour to the norm field and its warp to this signed vector field. Keep the contour fixed at 0–7.160033972903191. In particular, negative half-cycle displacement must not become a negative total-deformation magnitude.

4. For the accepted Python Result workflow on this installation, set Mechanical’s deformation multiplier to $3/7.160033972903191$. This is 1000 times the corresponding built-in Total Deformation setting, $0.003/7.160033972903191$, and produces the same arbitrary 3 mm visible peak. Verify this display-specific ratio when changing the software version or workflow; it does not multiply the reported eigenvector values. Fit the assembly once in the isometric view, increase the scene height by a factor of 1.3, then keep the camera fixed for all frames.

5. Test zero, positive-quarter-cycle and negative-quarter-cycle frames. Confirm opposite motion at the quarter cycles and a fixed color map. An identical-camera comparison against the built-in mode-20 plot differed at only 16 silhouette pixels out of 615023, at raster edges. The independent DPF/native-vector comparison at five phases gave relative L_2 error 2.53×10^{-8} . This comparison checks the displayed field; it does not establish physical displacement amplitude.
6. Evaluate each requested phase, check its audit, and export a 3840 by 2160 PNG with `ExtAPI.Graphics.ExportImage`. Keep the license active throughout result evaluation. Reject a frame if the workflow's recorded phase does not match the request. Preserve the camera, field audit, PNG hashes and native result identity.
7. Encode the 180 frames three times at 60 fps with the same nine-second duration and qualification as the default-mode clips. The source files and accepted export metadata are included under `reproduction/vid eos/`.

Encoding and acceptance

The executed macOS encoder uses hardware H.264 VideoToolbox, disallows a silent software fallback, requests an 80 Mbit/s target bitrate and a 120 Mbit/s maximum, uses 4:2:0 pixels, and writes a fast-start MP4. These are encoder settings, not a promise of a constant realized bitrate. Every final file must report 3840 by 2160, 60/1 fps, and 540 frames, and must decode completely without errors. Inspect the beginning, opposite phases and end for cropping, unreadable captions, changing camera position or stale result fields. The actual native PNGs and export audits, rather than a screenshot enlarged to 4K, establish the source resolution.